Data 602 E-Commerce Analysis

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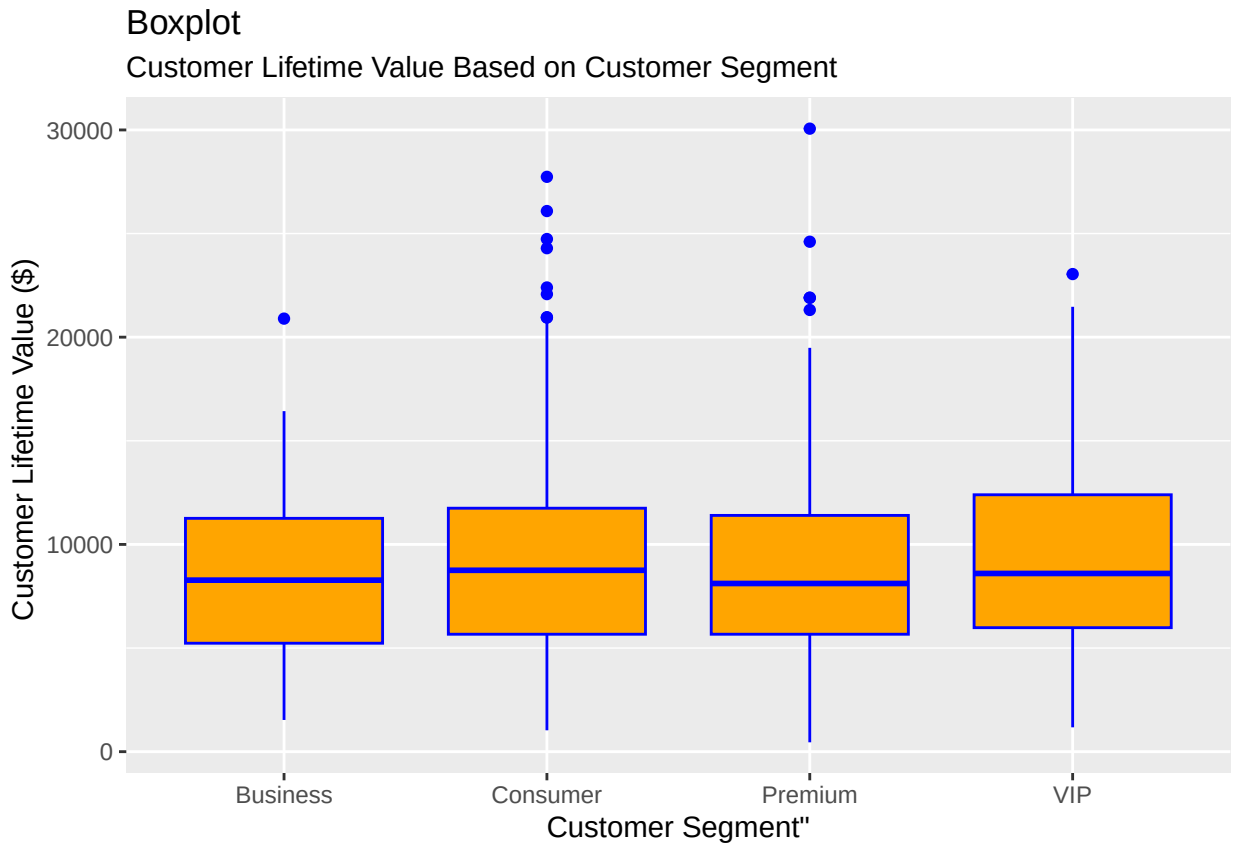
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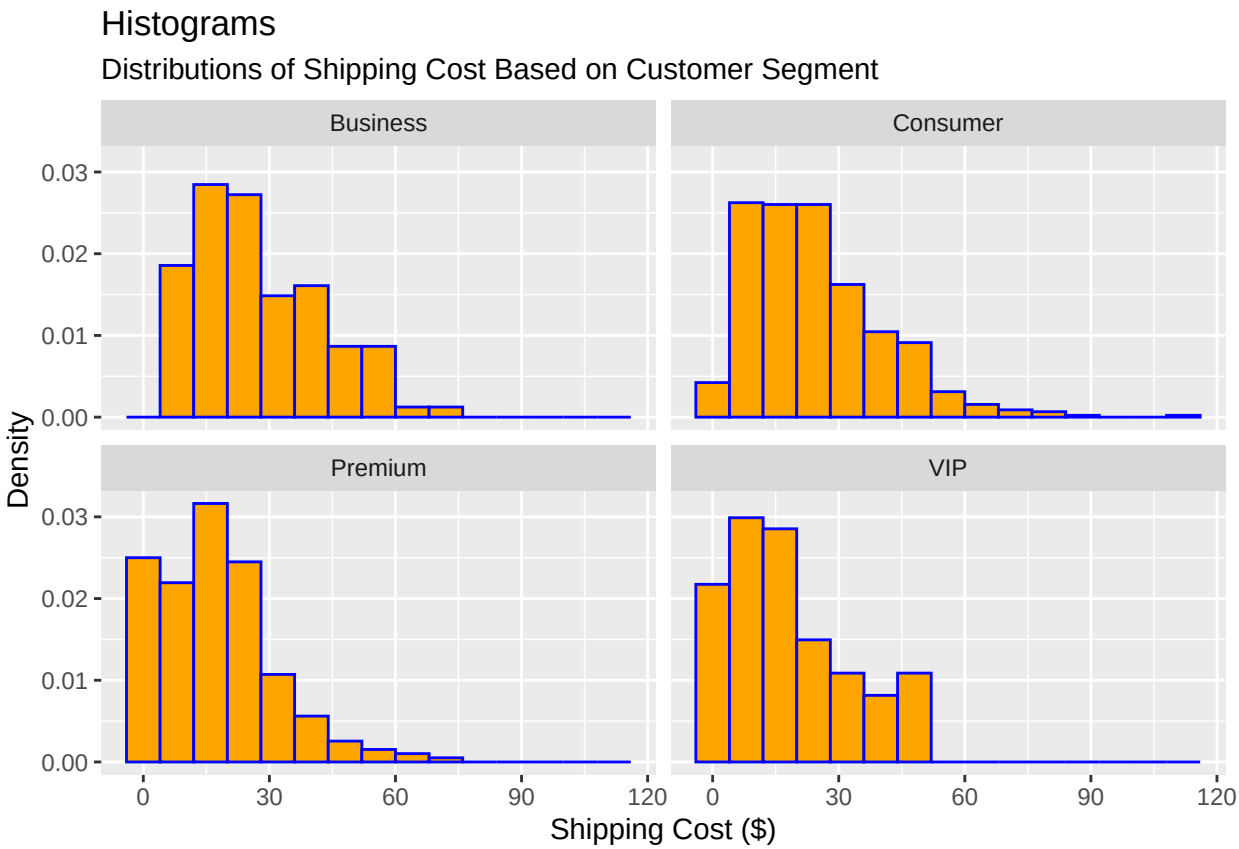
Exploratory Data Analysis



Based on the graph above, the customer segments have approximately the same median and spread of the customer lifetime value with the consumer segment having the most outliers.



The first 3 customer segments have similar medians of around 3.7 and similar spreads of data. VIP customers however have the highest median customer rating at around 3.9. The graph suggests that VIP customers tend to have slightly higher customer ratings on average and a higher first quartile than the other segments..



The plot shows that premium and VIP customers have more opportunities to receive shipping for free where as there are no opportunities for business and few for consumers.

Do Mean Customer Lifetime Values Differ Between The 4 Customer Segments?

(Each mu is of the customer lifetime value for the different segments of customers)

H0: u_business = u_consumer = u_premium = u_VIP

HA: At least one customer segment has a different mean lifetime value than at least one other customer segment's mean lifetime value

```
#Create linear regression model
lifetime_model = lm(customer_lifetime_value ~ customer_segment, data = sales_sample)

#Obtain P-value
anova(lifetime_model)[1,"Pr(>F)"]
```

```
## [1] 0.3165669
```

Since the p-value, 0.3166 is greater than 0.05, we fail to reject the null hypothesis that the mean customer lifetime value is equal across customer segments. Therefore there is insufficient evidence to conclude that the mean customer lifetime value differs between customer segments.

95% Confidence Intervals For Mean Customer Lifetime Value Separated by Customer Segment

Since we have the population data set, we know the population standard deviation of customer lifetime value based on customer segment. Therefore we can do a z-test instead of a t-test.

95% Confidence Interval of Mean Customer Lifetime Value for Business:

```
## [1] 7558.707 9252.582
```

95% Confidence Interval of Mean Customer Lifetime Value for Consumer:

```
## [1] 8758.257 9492.441
```

95% Confidence Interval of Mean Customer Lifetime Value for Premium:

```
## [1] 8284.138 9391.101
```

95% Confidence Interval of Mean Customer Lifetime Value for VIP:

```
## [1]  8556.186 10364.025
```

As we can see the 95% confidence intervals for mean customer lifetime value between customer segments overlap quite a bit. None of the intervals fall outside the range of any others. This supports the conclusion we came to of there being insufficient evidence of a difference in mean customer life time value between customer segments.

Do Mean Customer Ratings Differ Between Customer Segments

(Each mu is of the customer rating for the different segments of customers)

H0: u_business = u_consumer = u_premium = u_VIP

HA: At least one customer segment has a different mean customer rating than at least one other customer segment's mean customer rating

```
rating_model = lm(customer_rating ~ customer_segment, data = sales_sample)

anova(rating_model)[1,"Pr(>F)"]
```

```
## [1] 5.488825e-07
```

With a p-value of 0.000005489 which is less than 0.05, we have sufficient evidence to reject the null hypothesis that the mean customer rating is the same for all customer segments. This means that at least one customer segment's mean customer rating differs from at least one other customer segment's mean customer rating.

95% Confidence Interval For Mean Customer Rating For Each Customer Segment

95% Confidence Interval of Mean Customer Rating for Business:

```
## [1] 3.525970 3.713634
```

95% Confidence Interval of Mean Customer Rating for Consumer:

```
## [1] 3.570165 3.649408
```

95% Confidence Interval of Mean Customer Rating for Premium:

```
## [1] 3.691490 3.811368
```

95% Confidence Interval of Mean Customer Rating for VIP:

```
## [1] 3.767998 3.962436
```

The 95% confidence intervals for mean customer rating shows that premium customers' confidence interval falls outside the range of the consumer customers' confidence interval. The 95% confidence interval for mean customer rating for VIP customers also falls outside the range of the confidence intervals of both the business and the consumer customers. This supports our findings that at least one customer segment's mean customer rating differs from at least one other customer segment's.

Do Mean Shipping Costs Differ Between Customer Segments?

(Each mu is of the shipping costs for the different segments of customers)

H0: u_business = u_consumer = u_premium = u_VIP

HA: At least one customer segment has a different mean shipping cost than at least one other customer segment

```
shipping_model = lm(shipping_cost ~ customer_segment, data = sales_sample)

anova(shipping_model)[1,"Pr(>F)"]
```

```
## [1] 8.602353e-11
```

With a p-value of 0.00000000008602 which is less than 0.05, we have sufficient evidence to reject the null hypothesis that mean shipping cost is the same between customer segments. This means that at least one customer segment's mean shipping cost differs from at least one other customer segment's mean shipping cost.

95% Confidence Interval For Mean Shipping Cost For Each Customer Segment

95% Confidence Interval of Mean Shipping Cost for Business:

```
## [1] 24.00907 30.43251
```

95% Confidence Interval of Mean Shipping Cost for Consumer:

```
## [1] 22.84389 25.54732
```

95% Confidence Interval of Mean Shipping Cost for Premium:

```
## [1] 15.69347 19.16253
```

95% Confidence Interval of Mean Shipping Cost for VIP:

```
## [1] 15.37695 21.23501
```

Our 95% confidence intervals for mean shipping cost for the different customer segments supports our findings that at least one customer segment's mean shipping cost differs from at least one other customer segment's mean shipping cost. We can see that the premium and VIP customer segments' confidence intervals for mean shipping cost fall outside the confidence intervals for mean shipping cost of consumer and business customers.

Conclusion

We found there to be no difference in mean customer lifetime value between customer segments. We confirmed so using a linear regression model with customer lifetime value as the response variable and customer segment as a categorical predictor. We found evidence supporting this finding with our confidence intervals for mean customer lifetime value for each customer segment. Through the same process but with customer rating as the linear regression model's response variable and customer segment as a categorical predictor we found there to be at least one customer segment's mean customer rating that differs from at least one other customer segment's mean customer rating. We used confidence intervals for the customer rating for each customer segment to support our findings. We found the premium customers' confidence interval of mean customer rating to be outside the confidence interval of mean customer rating for consumer customers, and the VIP customers' confidence interval of mean customer rating to be outside the confidence interval of mean customer rating for both consumer and business customers. Through similar methods, we also found that the mean shipping cost of at least one customer segment differs from at least one other customer segment's mean shipping cost. With 95% confidence intervals we found evidence to support this finding. We found the premium and VIP customers' confidence interval for mean shipping cost to be below the range of the confidence intervals for mean shipping cost of consumer and business customers. Further analysis could be done to see if the increase in customer rating and decrease in shipping cost is due to a change in customer segment, or if a decrease in shipping cost is what causes an increase in customer rating.

Discussion

The goal of this analysis was to look at this E-commerce store's data and attempt to differentiate the habits of each customer segments. What we looked at was if there was a difference in spending between customers of a high ranked segment such as VIP and a lower ranked segment such as consumer, if there was a difference in ratings between customers of a higher ranked segment and customers of a lower ranked segment, and if there was a difference in shipping costs between customers of a higher ranked segment and customers of a lower ranked segment. Based on our findings in order to increase this store's ratings they could move customers of lower segments to higher segments, however doing so will not increase customer spending directly. This is valuable to the company because increasing a store's ratings tends to bring in more customers and therefore this could indirectly increase profit

Appendix

```
library(aod)
library(binom)
library(car)
library(collapsibleTree)
library(dbplyr)
library(dplyr)
library(Ecdat)
library(EnvStats)
library(faraway)
library(ggformula)
library(ggplot2)
library(ggpubr)
library(gmodels)
library(gridExtra)
library(htmltools)
library(ISLR)
library(knitr)
library(lawstat)
library(leaps)
library(lmtest)
library(MASS)
library(markdown)
library(mctest)
library(mdsr)
library(nnet)
#library(nycflights13)
library(nycflights23)
library(olsrr)
library(plyr)
library(purrr)
#library(plotly)
library(resampledata)
library(rmarkdown)
library(rpart)
library(rlang)
library(rpart.plot)
library(rvest)
library(SDaA)
library(shiny)
library(stringi)
library(tibble)
library(tidyr)
library(tidyselect)
library(tinytex)
library(yaml)
library(shiny)
library(aod)
library(ROCR)
library(pROC)
library(GGally)
library(mctest)
```

```
library(readxl)
library(mosaic)
library(mosaicData)
```

```
set.seed(30095675)
```

Import/Preparing Data

```
#Read the CSVs
customers = read.csv("customer_master-selected-columns.csv")
orders = read.csv("order_items.csv")
products = read.csv("product_catalog.csv")
sales = read.csv("ecommerce_sales_customer_analytics_150k.csv")
```

```
#Create data frame with features of interest
sales_pop = data.frame(sales[,c("customer_segment","customer_rating","customer_age","gender","customer_
```

```
#Get rid of rows with null values from the population dataset
sales_pop = na.omit(sales_pop)

#Turning catagorical features into factors to be able to use in calculations
sales_pop$customer_segment = as.factor(sales_pop$customer_segment)
sales_pop$gender = as.factor(sales_pop$gender)
sales_pop$male <- ifelse(sales_pop$gender == "Male", 1, 0)

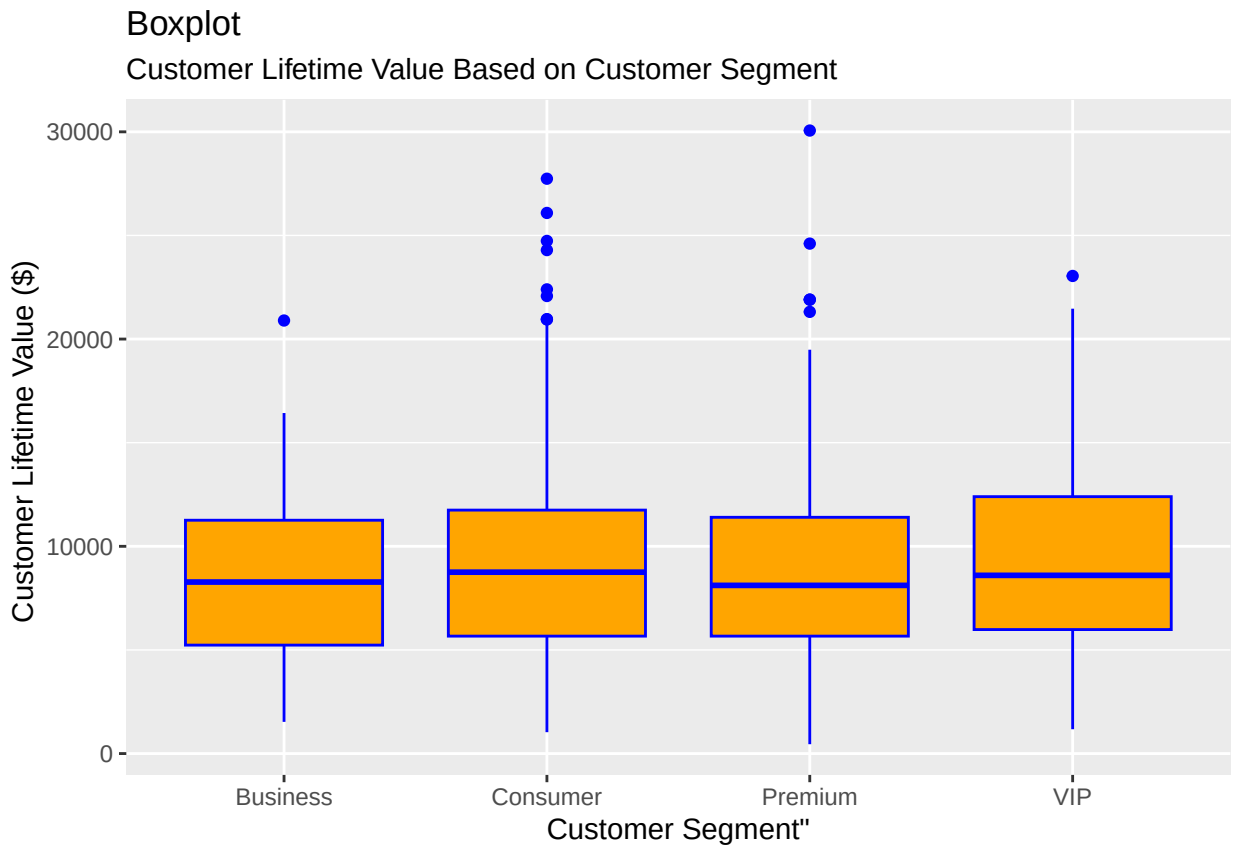

#Take a sample of n = 1000 customer transactions
sales_sample = sales_pop[sample(nrow(sales_pop), size =1000, replace = FALSE),]
```

Exploratory Data Analysis

```
#Boxplot
ggplot(sales_sample,
       aes(x = customer_segment,
           y = customer_lifetime_value)) +

  geom_boxplot(col = 'blue',
               fill = 'orange') +

  xlab('Customer Segment"') +
  ylab("Customer Lifetime Value ($) ") +
  ggtitle("Boxplot",
          subtitle = "Customer Lifetime Value Based on Customer Segment")
```




```
#Boxplot
ggplot(sales_sample,
       aes(x = customer_segment,
           y = customer_rating)) +

  geom_boxplot(col = 'blue',
               fill = 'orange') +

  xlab('Customer Segment') +
  ylab("Customer Rating") +
  ggtitle("Boxplot",
          subtitle = "Customer Rating Based on Customer Segment")
```

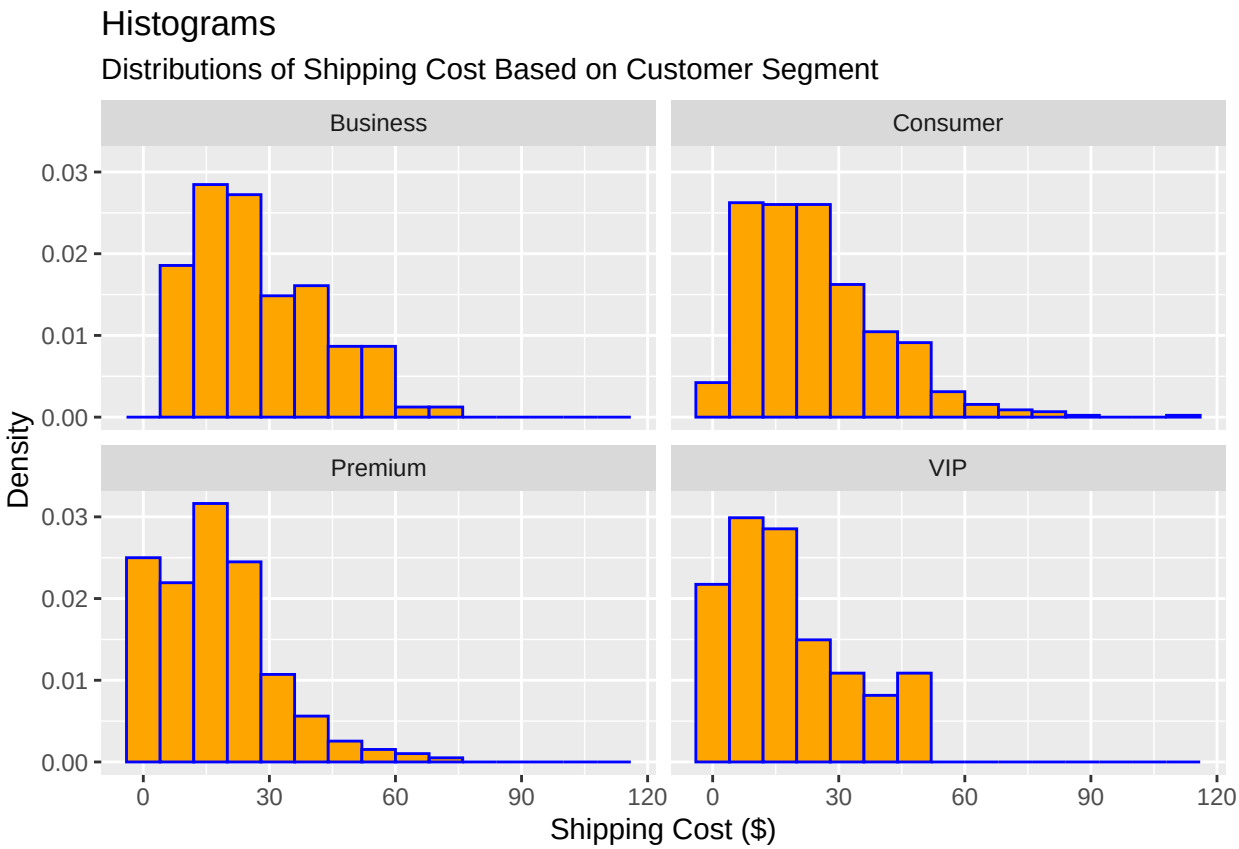



```
#Histogram
ggplot(sales_sample,
       aes(x = shipping_cost,
           y = after_stat(density))) +

  geom_histogram(col = 'blue',
                 fill = 'orange',
                 binwidth = 8) +

  facet_wrap(~ customer_segment) +

  xlab('Shipping Cost ($)') +
  ylab("Density") +
  ggtitle("Histograms",
          subtitle = "Distributions of Shipping Cost Based on Customer Segment")
```



Do Mean Customer Lifetime Values Differ Between The 4 Customer Segments?

```
#Create linear regression model
lifetime_model = lm(customer_lifetime_value ~ customer_segment, data = sales_sample)

#Obtain P-value
anova(lifetime_model)[1,"Pr(>F)"]
```

```
## [1] 0.3165669
```

95% Confidence Intervals For Mean Customer Lifetime Value Separated by Customer Segment

```
sigma_b = sd(sales_pop$customer_lifetime_value[sales_pop$customer_segment == "Business"])

business = sales_sample$customer_lifetime_value[sales_sample$customer_segment == "Business"]

mean(business) +c(-1,1)*qnorm(0.975)*(sigma_b/sqrt(length(business)))
```

```
## [1] 7558.707 9252.582
```

```
sigma_c = sd(sales_pop$customer_lifetime_value[sales_pop$customer_segment == "Consumer"])

consumer = sales_sample$customer_lifetime_value[sales_sample$customer_segment == "Consumer"]

mean(consumer) +c(-1,1)*qnorm(0.975)*(sigma_c/sqrt(length(consumer)))
```

```
## [1] 8758.257 9492.441
```

```
sigma_p = sd(sales_pop$customer_lifetime_value[sales_pop$customer_segment == "Premium"])

premium = sales_sample$customer_lifetime_value[sales_sample$customer_segment == "Premium"]

mean(premium) +c(-1,1)*qnorm(0.975)*(sigma_p/sqrt(length(premium)))
```

```
## [1] 8284.138 9391.101
```

```
sigma_v = sd(sales_pop$customer_lifetime_value[sales_pop$customer_segment == "VIP"])

vip = sales_sample$customer_lifetime_value[sales_sample$customer_segment == "VIP"]

mean(vip) +c(-1,1)*qnorm(0.975)*(sigma_v/sqrt(length(vip)))
```

```
## [1]  8556.186 10364.025
```

Do Mean Customer Ratings Differ Between Customer Segments

```
rating_model = lm(customer_rating ~ customer_segment, data = sales_sample)

anova(rating_model)[1,"Pr(>F)"]
```

```
## [1] 5.488825e-07
```

95% Confidence Interval For Mean Customer Rating For Each Customer Segment

```
sigma_b_r = sd(sales_pop$customer_rating[sales_pop$customer_segment == "Business"])

business_r = sales_sample$customer_rating[sales_sample$customer_segment == "Business"]

mean(business_r) +c(-1,1)*qnorm(0.975)*(sigma_b_r/sqrt(length(business_r)))
```

```
## [1] 3.525970 3.713634
```

```r
sigma_c_r = sd(sales_pop$customer_rating[sales_pop$customer_segment == "Consumer"])

consumer_r = sales_sample$customer_rating[sales_sample$customer_segment == "Consumer"]

mean(consumer_r) +c(-1,1)*qnorm(0.975)*(sigma_c_r/sqrt(length(consumer_r)))
```

```
## [1] 3.570165 3.649408
```

```r
sigma_p_r = sd(sales_pop$customer_rating[sales_pop$customer_segment == "Premium"])

premium_r = sales_sample$customer_rating[sales_sample$customer_segment == "Premium"]

mean(premium_r) +c(-1,1)*qnorm(0.975)*(sigma_p_r/sqrt(length(premium_r)))
```

```
## [1] 3.691490 3.811368
```

```r
sigma_v_r = sd(sales_pop$customer_rating[sales_pop$customer_segment == "VIP"])

vip_r = sales_sample$customer_rating[sales_sample$customer_segment == "VIP"]

mean(vip_r) +c(-1,1)*qnorm(0.975)*(sigma_v_r/sqrt(length(vip_r)))
```

```
## [1] 3.767998 3.962436
```

Do Mean Shipping Costs Differ Between Customer Segments?

```r
shipping_model = lm(shipping_cost ~ customer_segment, data = sales_sample)

anova(shipping_model)[1,"Pr(>F)"]
```

```
## [1] 8.602353e-11
```

95% Confidence Interval For Mean Shipping Cost For Each Customer Segment

```r
sigma_b_s = sd(sales_pop$shipping_cost[sales_pop$customer_segment == "Business"])

business_s = sales_sample$shipping_cost[sales_sample$customer_segment == "Business"]

mean(business_s) +c(-1,1)*qnorm(0.975)*(sigma_b_s/sqrt(length(business_s)))
```

```
## [1] 24.00907 30.43251
```

```r
sigma_c_s = sd(sales_pop$shipping_cost[sales_pop$customer_segment == "Consumer"])

consumer_s = sales_sample$shipping_cost[sales_sample$customer_segment == "Consumer"]

mean(consumer_s) +c(-1,1)*qnorm(0.975)*(sigma_c_s/sqrt(length(consumer_s)))
```

```
## [1] 22.84389 25.54732
```

```r
sigma_p_s = sd(sales_pop$shipping_cost[sales_pop$customer_segment == "Premium"])

premium_s = sales_sample$shipping_cost[sales_sample$customer_segment == "Premium"]

mean(premium_s) +c(-1,1)*qnorm(0.975)*(sigma_p_s/sqrt(length(premium_s)))
```

```
## [1] 15.69347 19.16253
```

```r
sigma_v_s = sd(sales_pop$shipping_cost[sales_pop$customer_segment == "VIP"])

vip_s = sales_sample$shipping_cost[sales_sample$customer_segment == "VIP"]

mean(vip_s) +c(-1,1)*qnorm(0.975)*(sigma_v_s/sqrt(length(vip_s)))
```

```
## [1] 15.37695 21.23501
```